Pima Indian Diabetes Prediction

End-to-End Data Science & Machine Learning Project

Predicting diabetes (Outcome) from diagnostic features — Pregnancies, Glucose, BloodPressure, SkinThickness, Insulin, BMI, DiabetesPedigreeFunction, Age

Source notebook:	Diabetes Prediction.ipynb
Cells:	75 (markdown + code with saved outputs)
Dataset:	diabetes.csv • 768 rows × 9 columns
Models compared:	Logistic Regression • Random Forest • SVM
Export date:	September 30, 2026

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Pima Indian Diabetes Prediction - Data Science Project

This is an end-to-end **Data Science and Machine Learning** project to build a predictive system that determines whether a patient has diabetes or not based on medical diagnostic factors such as Glucose Level, Blood Pressure, Skin Thickness, Insulin Level, BMI, Diabetes Pedigree Function, Age, and Pregnancies.

Unlike a pure data analysis project, this project goes beyond EDA and visualization to include data preprocessing, feature engineering, model training, evaluation, and selection of the best classification model for diabetes prediction.

About the Dataset

This dataset is originally from the National Institute of Diabetes and Digestive and Kidney Diseases. The objective of the dataset is to diagnostically predict whether or not a patient has diabetes, based on certain diagnostic measurements included in the dataset. Several constraints were placed on the selection of these instances from a larger database. In particular, all patients here are females at least 21 years old of Pima Indian heritage.

The datasets consists of several medical predictor variables and one target variable, Outcome. Predictor variables includes the number of pregnancies the patient has had, their BMI, insulin level, age, and so on.

Problem Statement

Diabetes is a major chronic disease with high prevalence in the Pima Indian population, and delayed diagnosis leads to severe complications like heart disease, kidney failure, and vision loss. Manual diagnosis based on multiple medical tests is time-consuming and subject to human interpretation.

The problem is to develop a **data-driven machine learning classification system** that can automatically predict whether a patient has diabetes (Outcome = 1) or not (Outcome = 0) using 8 diagnostic features:

Pregnancies, Glucose, BloodPressure, SkinThickness, Insulin, BMI, DiabetesPedigreeFunction, and Age.

This Data Science project solves this by following a complete ML workflow: exploratory analysis, handling invalid zeros/missing values, feature scaling, handling class imbalance (65% Negative vs 35% Positive), train-test splitting, training multiple classifiers, and selecting the best model based on Accuracy, Precision, Recall, F1-Score, and ROC-AUC for reliable early prediction.

Objective (Data Science Project)

The objective is to build, evaluate, and select the best machine learning model for diabetes prediction:

1. **Data Understanding & EDA:** Explore distributions, correlations, class balance, and relationship of each feature with `Outcome`.
2. **Data Preprocessing:** Handle biologically invalid zeros (Glucose, BloodPressure, SkinThickness, Insulin, BMI), handle outliers, encode target, and apply feature scaling (StandardScaler / MinMaxScaler).
3. **Feature Engineering & Selection:** Analyze feature importance (correlation, model-based importance) and select relevant predictors.
4. **Model Building:** Split data into train-test sets (e.g., 80/20, stratified) and train multiple classifiers such as Logistic Regression, KNN, SVM, Decision Tree, Random Forest, and XGBoost.
5. **Model Evaluation & Selection:** Compare models using Accuracy, Precision, Recall, F1-Score, Confusion Matrix, and ROC-AUC with cross-validation, tune hyperparameters, and finalize the best model for deployment-ready prediction.

Research Questions (Data Science / ML Focus)

This project seeks to answer the following modelling-oriented questions:

1. How should invalid zeros and outliers in Glucose, BloodPressure, SkinThickness, Insulin, and BMI be handled to improve model performance?
2. Which features contribute most to diabetes prediction, and can feature selection improve accuracy and reduce overfitting?
3. Which classification algorithm (Logistic Regression vs KNN vs SVM vs Random Forest vs XGBoost) gives the best Precision, Recall, F1-Score, and ROC-AUC on unseen test data?
4. How does class imbalance (65% Non-Diabetic vs 35% Diabetic) affect model performance, and do techniques like Stratified Split, SMOTE, or class_weight improve Recall for diabetic patients?
5. What is the final best model and its generalization performance, and how can it be used for real-time prediction on new patient data?

Step 1: Import libraries

In [1] — Code

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
warnings.filterwarnings('ignore')
```

Step 2 - Loading the Data

In [2] — Code

```
df=pd.read_csv('diabetes.csv')
```

Step 3 - Exploratory Data Analysis (EDA)

In [3] — Code

```
df.head()
```

Output:

```
   Pregnancies  Glucose  BloodPressure  SkinThickness  Insulin   BMI  \
0            6      148           72.0             35    169.5  33.6
1            1       85           66.0             29    102.5  26.6
2            8      183           64.0             32    169.5  23.3
3            1       89           66.0             23     94.0  28.1
4            0      137           40.0             35    168.0  43.1

   DiabetesPedigreeFunction  Age  Outcome
0                     0.627   50        1
1                     0.351   31        0
2                     0.672   32        1
3                     0.167   21        0
4                     2.288   33        1
```

In [4] — Code

```
df.shape
```

Output:

```
(768, 9)
```

Step 4 - Starting Data Cleaning

In [5] — Code

```
df.isnull().sum()
```

Output:

```
Pregnancies                 0
Glucose                     0
BloodPressure               0
SkinThickness               0
Insulin                     0
BMI                         0
DiabetesPedigreeFunction    0
Age                         0
Outcome                     0
dtype: int64
```

Step 5 - Check for Duplicates

In [6] — Code

```
df[df.duplicated()]
```

Output:

```
Empty DataFrame
Columns: [Pregnancies, Glucose, BloodPressure, SkinThickness, Insulin, BMI, DiabetesPedigreeFunction, Age, Outcome]
Index: []
```

In [7] — Code

```
(df == 0).sum()
```

Output:

```
Pregnancies                 111
Glucose                       0
BloodPressure                 0
SkinThickness                 0
Insulin                       0
BMI                           0
DiabetesPedigreeFunction      0
Age                           0
Outcome                     500
dtype: int64
```

In [8] — Code

```
df.describe()
```

Output:

```
       Pregnancies     Glucose  BloodPressure  SkinThickness     Insulin  \
```

```
count   768.000000  768.000000     768.000000     768.000000  768.000000
mean      3.845052  121.677083      72.389323      29.089844  141.753906
std       3.369578   30.464161      12.106039       8.890820   89.100847
min       0.000000   44.000000      24.000000       7.000000   14.000000
25%       1.000000   99.750000      64.000000      25.000000  102.500000
50%       3.000000  117.000000      72.000000      28.000000  102.500000
75%       6.000000  140.250000      80.000000      32.000000  169.500000
max      17.000000  199.000000     122.000000      99.000000  846.000000

              BMI  DiabetesPedigreeFunction         Age     Outcome
count  768.000000                768.000000  768.000000  768.000000
mean    32.434635                  0.471876   33.240885    0.348958
std      6.880498                  0.331329   11.760232    0.476951
min     18.200000                  0.078000   21.000000    0.000000
25%     27.500000                  0.243750   24.000000    0.000000
50%     32.050000                  0.372500   29.000000    0.000000
75%     36.600000                  0.626250   41.000000    1.000000
max     67.100000                  2.420000   81.000000    1.000000
```

Step 6 - Checking for Invalid Categories

In [9] — Code

```
df['Pregnancies'].value_counts(normalize=True)
```

Output:

```
Pregnancies
1     0.175781
0     0.144531
2     0.134115
3     0.097656
4     0.088542
5     0.074219
6     0.065104
7     0.058594
8     0.049479
9     0.036458
10    0.031250
11    0.014323
13    0.013021
12    0.011719
14    0.002604
15    0.001302
17    0.001302
Name: proportion, dtype: float64
```

Step 7 - Make outcomes human-readable

In [10] — Code

```
df['Outcome'] = df['Outcome'].replace({
 0: 'Negative',
 1: 'Positive'
})
```

In [11] — Code

```
df.head()
```

Output:

```
   Pregnancies  Glucose  BloodPressure  SkinThickness  Insulin   BMI  \
0            6      148           72.0             35    169.5  33.6
1            1       85           66.0             29    102.5  26.6
2            8      183           64.0             32    169.5  23.3
3            1       89           66.0             23     94.0  28.1
4            0      137           40.0             35    168.0  43.1

   DiabetesPedigreeFunction  Age   Outcome
0                     0.627   50  Positive
1                     0.351   31  Negative
2                     0.672   32  Positive
3                     0.167   21  Negative
4                     2.288   33  Positive
```

In [12] — Code

```
df.info()
```

Output:

```
&lt;class 'pandas.DataFrame'&gt;
RangeIndex: 768 entries, 0 to 767
Data columns (total 9 columns):
 #   Column                    Non-Null Count  Dtype
---  ------                    --------------  -----
 0   Pregnancies               768 non-null    int64
 1   Glucose                   768 non-null    int64
 2   BloodPressure             768 non-null    float64
 3   SkinThickness             768 non-null    int64
 4   Insulin                   768 non-null    float64
 5   BMI                       768 non-null    float64
 6   DiabetesPedigreeFunction  768 non-null    float64
 7   Age                       768 non-null    int64
 8   Outcome                   768 non-null    object
dtypes: float64(4), int64(4), object(1)
memory usage: 54.1+ KB
```

Step 8 - Exploratory Data Analysis

In the exploratory data analysis, I will be looking at the distribution of the data, the correlation between the features, and the relationship between the features and the target variable. I will start by looking at the distribution of the data, followed by relationship between the target variable and independent variables.

In [13] — Code

```
plt.figure(figsize=(5,5))
plt.pie(df['Outcome'].value_counts(), labels=['Negative', 'Positive'], autopct='%1.1f%%', shadow=False,
startangle=90,colors=['#228B22','#DC143C'])
plt.title('Diabetes Outcome')
```

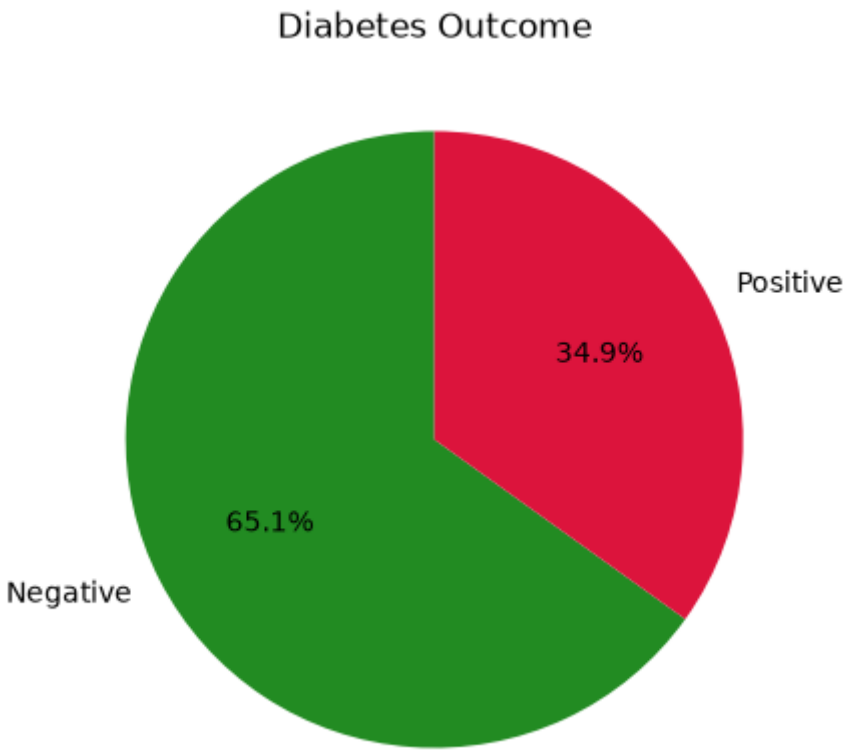


Figure — output of cell [13]

Age Distribution and Diabetes

In [14] — Code

```
sns.catplot(x="Outcome", y="Age", kind="swarm", data=df, palette=['#DC143C','#228B22'])
```

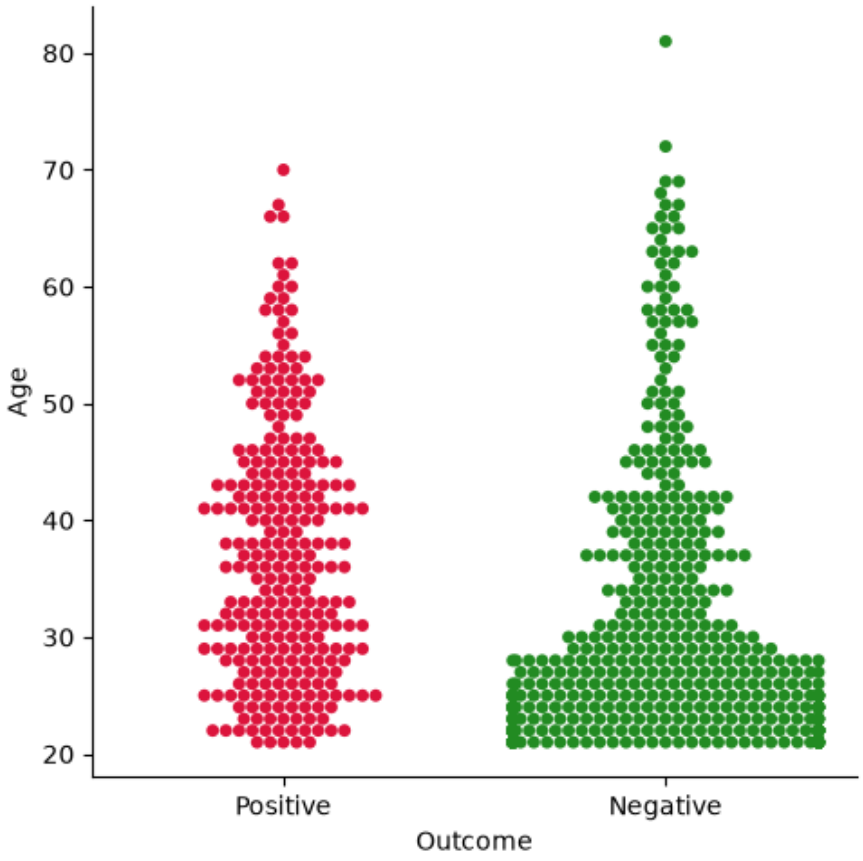


Figure — output of cell [14]

From the graph, it is quite clear that majority of the patients are adult within the age group of 20-30 years. Patients in the age range 40-55 years are more prone to diabetes, as compared to other age groups. Since the number adults in the age group 20-30 years is more, the number of patients with diabetes is also more as compared of other age groups.

Pregnancies and Diabetes

In [15] — Code

```
plt.figure(figsize=(15, 5))

plt.subplot(1, 2, 1)
plt.title('pregnancies increased risk of diabetes.')
sns.barplot(x='Outcome', y='Pregnancies', data=df,palette=['#DC143C','#228B22'])

plt.subplot(1, 2, 2)
plt.title('pregnancies increased risk of diabetes.')
sns.boxplot(x='Outcome', y='Pregnancies', data=df,palette=['#DC143C','#228B22'])

plt.tight_layout()
```

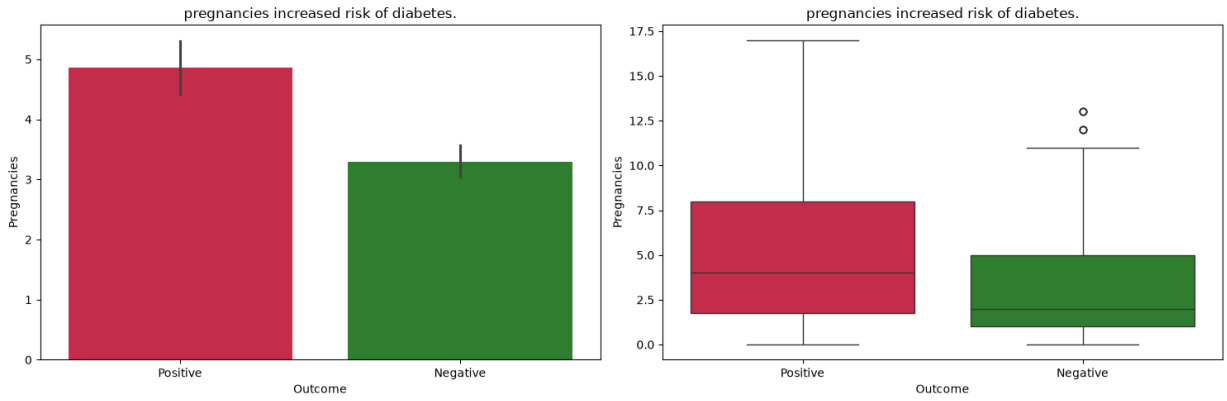


Figure — output of cell [15]

Both boxplot and barplot shows strange relation between the number of preganacies and diabetes. According to the graphs the increased number of pregnancies highlights increased risk of diabetes.

Glucose vs Diabetes

In [16] — Code

```
sns.boxplot(x='Outcome', y='Glucose', data=df,palette=['#DC143C','#228B22']).set_title('Glucose vs Diabe
```

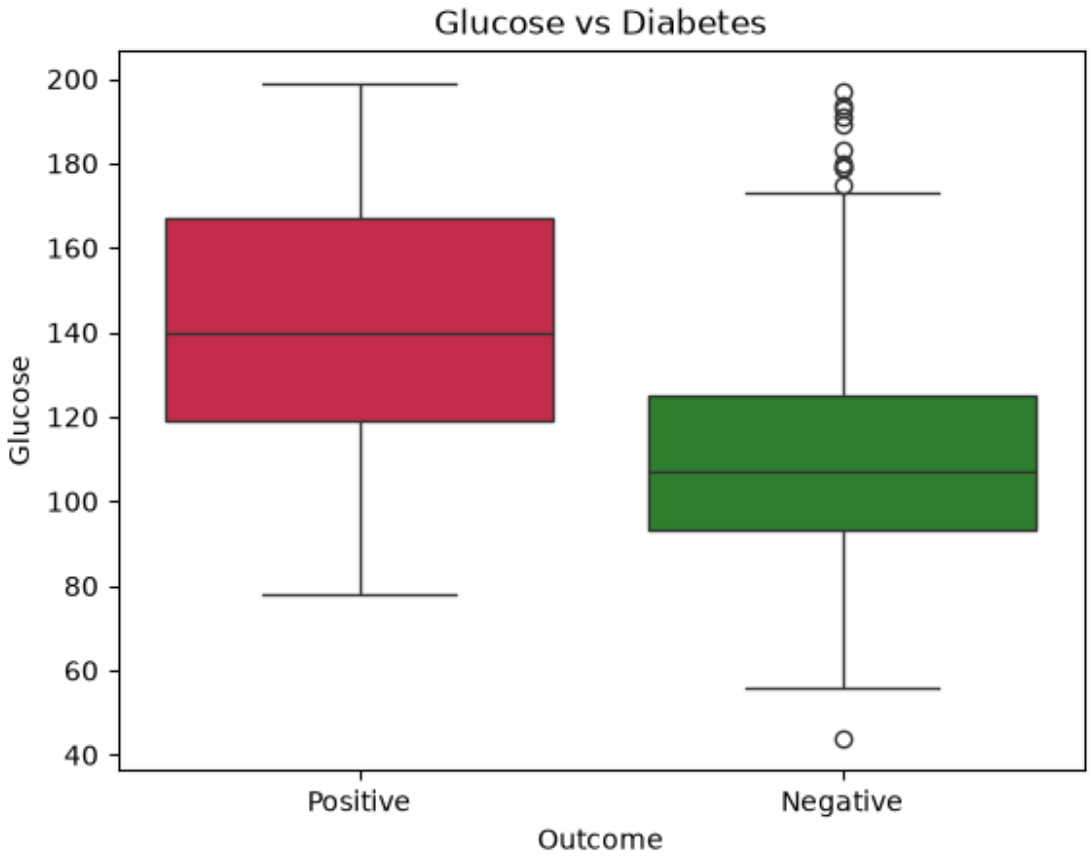


Figure — output of cell [16]

Glucose level plays a major role in determine whether the patient is diabetic or not. The patients with median gluocse level less than 120 are more likely to be non-diabetic. The patients with median gluocse level greather than 140 are more likely to be diabetic. Therefore, high gluocose levels is a good indicator of diabetes.

Blood Pressure VS Diabetes

In [17] — Code

```
plt.figure(figsize=(15, 5))

plt.subplot(1, 2, 1)
plt.title('Blood Pressure VS Diabetes')
sns.barplot(x='Outcome', y='BloodPressure', data=df,palette=['#DC143C','#228B22'])

plt.subplot(1, 2, 2)
plt.title('Blood Pressure VS Diabetes')
sns.boxplot(x='Outcome', y='BloodPressure', data=df,palette=['#DC143C','#228B22'])

plt.tight_layout()
```

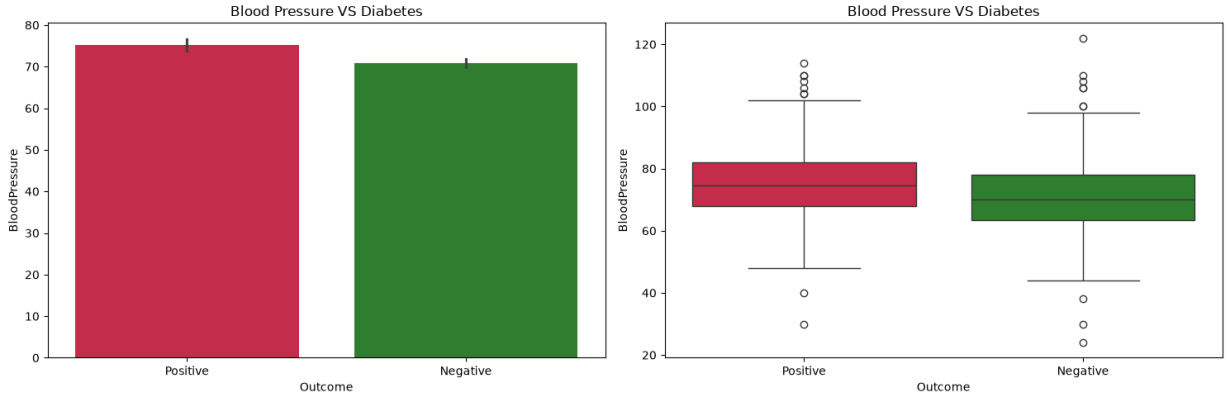


Figure — output of cell [17]

Both the boxplot and Barplot provides clear understanding of the realtion between the blood pressure and diabetes. The boxplot shows that the median of the blood pressure for the diabetic patients is slightly higher than the non-diabetic patients. The voilinplot shows that the distribution of the blood pressure for the diabetic patients is slightly higher than the non-diabetic patients. But there has been not enough evidence to conclude that the blood pressure is a good predictor of diabetes.

Skin Thickness VS Diabetes

In [18] — Code

```
plt.figure(figsize=(15, 5))

plt.subplot(1, 2, 1)
plt.title('Skin Thickness VS Diabetes')
sns.barplot(x='Outcome', y='SkinThickness', data=df,palette=['#DC143C','#228B22'])

plt.subplot(1, 2, 2)
plt.title('Skin Thickness VS Diabetes')
sns.boxplot(x='Outcome', y='SkinThickness', data=df,palette=['#DC143C','#228B22'])

plt.tight_layout()
```

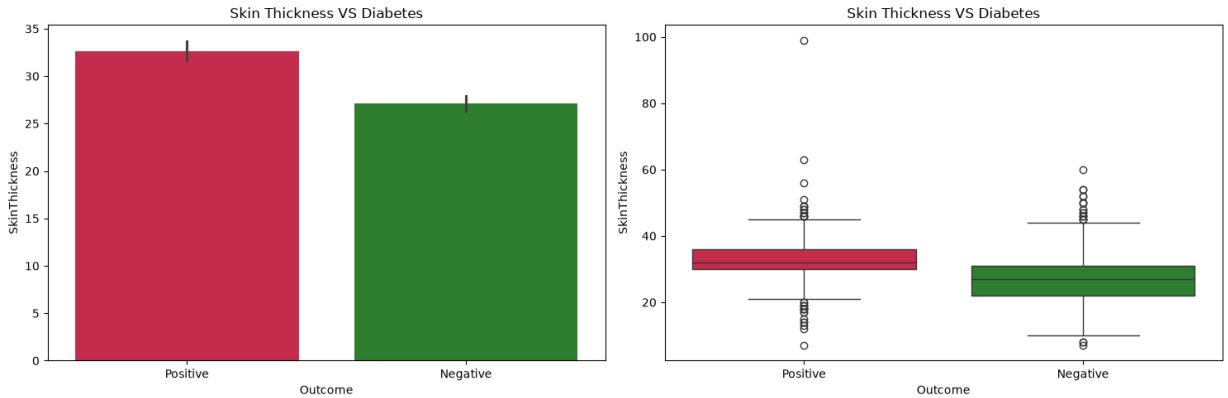


Figure — output of cell [18]

Here both the boxplot and violinplot reveals the effect of diabetes on skin thickness. As obserevd in the boxplot, the median of skin thickness is higher for the diabetic patients than the non-diabetic patients, where non diabetic patients have median skin thickness near 20 in comparison to skin thickness nearly 30 in diabetic patients. The voilinpplot shows the distribution of patients' skin thickness amoung the patients, where the non diabetic ones have greater distribution near 20 and diabetic much less distribution near 20 and increased distribution near 30. Therefore, skin thickness can be a indicator of diabetes.

Insulin VS Diabetes

In [19] — Code

```
plt.figure(figsize=(15, 5))

plt.subplot(1, 2, 1)
plt.title('Insulin VS Diabetes')
sns.barplot(x='Outcome', y='Insulin', data=df,palette=['#DC143C','#228B22'])

plt.subplot(1, 2, 2)
plt.title('Insulin VS Diabetes')
sns.violinplot(x='Outcome', y='Insulin', data=df,palette=['#DC143C','#228B22'])

plt.tight_layout()
```

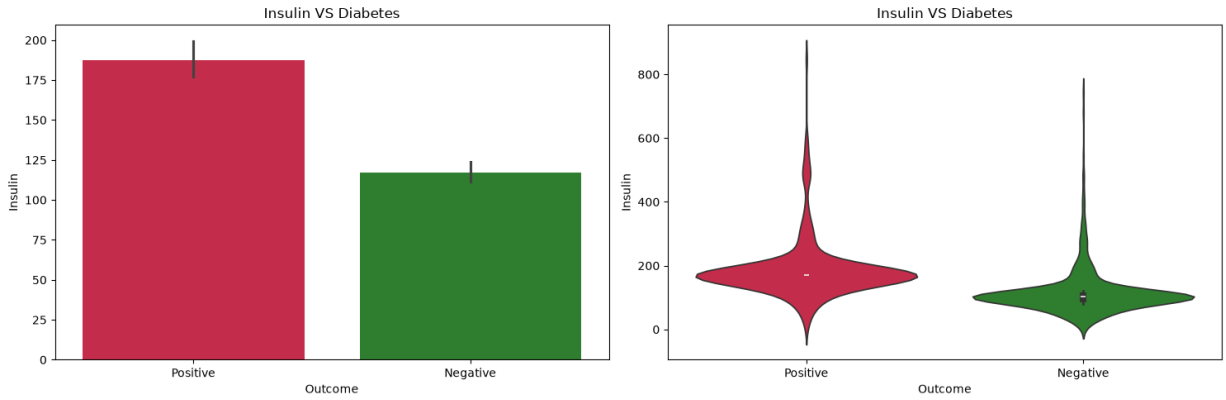


Figure — output of cell [19]

Insulin is a major body hormone that regulates glucose metabolism. Insulin is required for the body to efficiently use sugars, fats and proteins. Any change in insulin amount in the body would result in change glucose levels as well. Here the boxplot and violinplot shows the distribution of insulin level in patients. In non diabetic patients the insulin level is near to 100, whereas in diabetic patients the insulin level is near to 200. In the voilinplot we can see that the distribution of insulin level in non diabetic patients is more spread out near 100, whereas in diabetic patients the distribution is contracted and shows a little bit spread in higher insulin levels. This shows that the insulin level is a good indicator of diabetes.

BMI VS Diabetes

In [20] — Code

```
plt.figure(figsize=(15, 5))

plt.subplot(1, 2, 1)
plt.title('BMI VS Diabetes')
sns.violinplot(x='Outcome', y='BMI', data=df,palette=['#DC143C','#228B22'])

plt.subplot(1, 2, 2)
plt.title('BMI VS Diabetes')
sns.boxplot(x='Outcome', y='BMI', data=df,palette=['#DC143C','#228B22'])

plt.tight_layout()
```

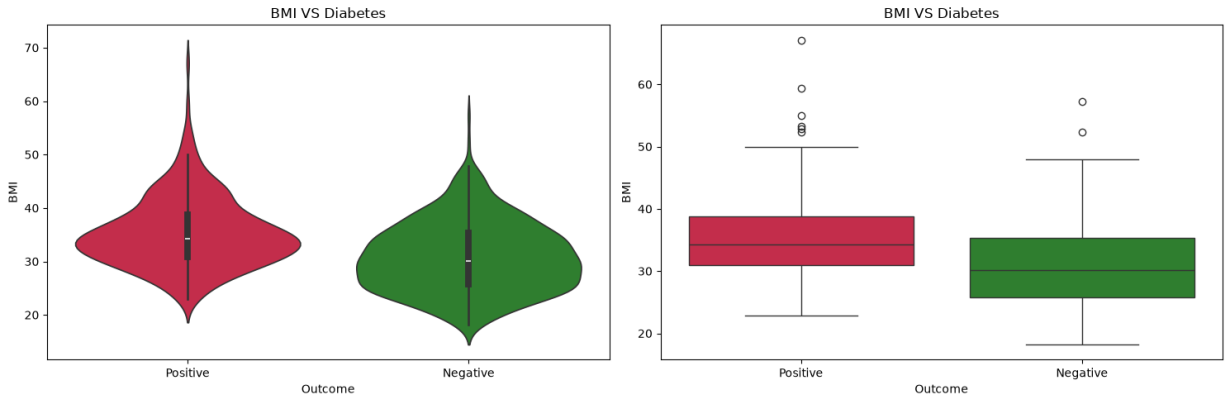


Figure — output of cell [20]

Both graphs highlights the role of BMI in diabetes prediction. Non diabetic patients have a normal BMI within the range of 25-35 whereas the diabetic patients have a BMI greater than 35. The violinplot reveals the BMI distribution, where the non dibetic patients have a increased spread from 25 to 35 with narrows after 35. However in diabetic patients there is increased spread at 35 and increased spread 45-50 as compared to non diabetic patients.Therefore BMI is a good predictor of diabetes and obese people are more likely to be diabetic.

Diabetes Pedigree Function VS Diabetes

In [21] — Code

```
plt.figure(figsize=(15, 5))

plt.subplot(1, 2, 1)
plt.title('Diabetes Pedigree Function VS Diabetes')
sns.violinplot(x='Outcome', y='DiabetesPedigreeFunction', data=df,palette=['#DC143C','#228B22'])

plt.subplot(1, 2, 2)
plt.title('BMI VS Diabetes')
sns.boxplot(x='Outcome', y='DiabetesPedigreeFunction', data=df,palette=['#DC143C','#228B22'])

plt.tight_layout()
```

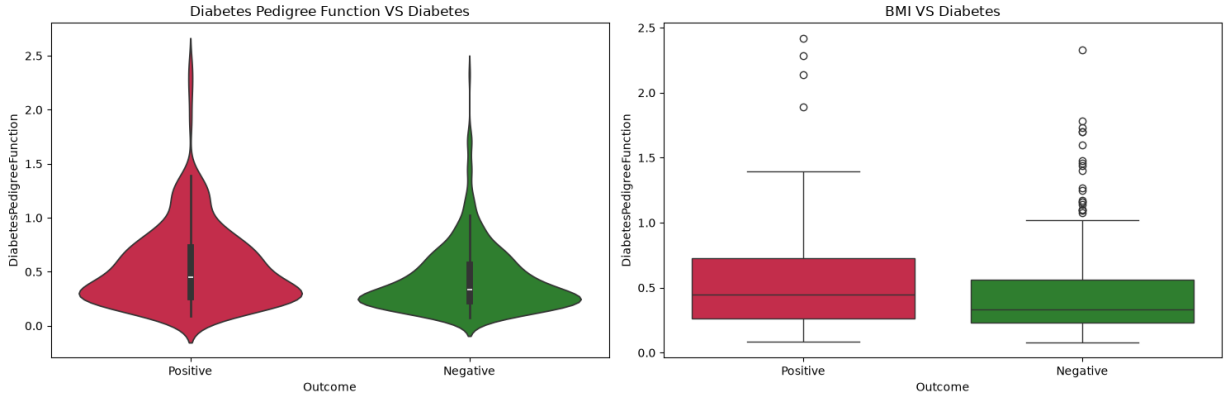


Figure — output of cell [21]

Diabetes Pedigree Function (DPF) calculates diabetes likelihood depending on the subject's age and his/her diabetic family history. From the boxplot, the patients with lower DPF, are much less likely to have diabetes. The patients with higher DPF, are much more likely to have diabetes. In the violinplot, majority of the non diabetic patients have a DPF of 0.25-0.35, whereas the diabetic patients have a increased DPF, which is shown by the their distribution in the violinplot where there is a increased spread in the DPF from 0.5 -1.5. Therefore the DPF is a good indicator of diabetes.

Diabetes Prediction Models

For predictiong the diabetes, I will be using the following algorithms:

1. Logistic Regression
2. Random Forest Classifier
3. Support Vector Machine

In [22] — Code

```
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score
```

In [23] — Code

```
X_train, X_test, y_train, y_test = train_test_split(df.drop('Outcome',axis=1),df['Outcome'],test_size=0.
_state=42)
```

Logistic Regression

In [24] — Code

```
from sklearn.linear_model import LogisticRegression
```

In [25] — Code

```
lr = LogisticRegression()
```

In [26] — Code

```
lr.fit(X_train,y_train)
```

Output:

```
LogisticRegression()
```

In [27] — Code

```
prediction = lr.predict(X_test)
```

In [28] — Code

```
actual = y_test.values
```

In [29] — Code

```
lr_acc = accuracy_score(actual, prediction)
print(f"Logistic Regression Accuracy: {lr_acc*100:.2f}%")
print(f"Correct: {int(lr_acc*len(actual))} / {len(actual)}")
```

Output:

```
Logistic Regression Accuracy: 76.62%
Correct: 118 / 154
```

Random Forest Classifier

In [30] — Code

```
from sklearn.ensemble import RandomForestClassifier
```

In [31] — Code

```
rfc = RandomForestClassifier(n_estimators=100,random_state=42)
```

In [32] — Code

```
rfc.fit(X_train, y_train)
```

Output:

```
RandomForestClassifier(random_state=42)
```

In [33] — Code

```
prediction = rfc.predict(X_test)
```

In [34] — Code

```
rfc_acc = accuracy_score(actual, prediction)
print(f"Random Forest classifier Accuracy: {rfc_acc*100:.2f}%")
print(f"Correct: {int(rfc_acc*len(actual))} / {len(actual)}")
```

Output:

```
Random Forest classifier Accuracy: 88.31%
Correct: 136 / 154
```

Support Vector Machine

In [35] — Code

```
from sklearn.svm import SVC
```

In [36] — Code

```
svm = SVC(kernel='linear', random_state=0)
```

In [37] — Code

```
svm.fit(X_train, y_train)
```

Output:

```
SVC(kernel='linear', random_state=0)
```

In [38] — Code

```
prediction = svm.predict(X_test)
```

In [39] — Code

```
svm_acc = accuracy_score(actual, prediction)
print(f"Support Vector Machine Accuracy: {svm_acc*100:.2f}%")
print(f"Correct: {int(svm_acc*len(actual))} / {len(actual)}")
```

Output:

```
Support Vector Machine Accuracy: 79.22%
Correct: 122 / 154
```

Comparing the models

In [40] — Code

```
sns.barplot(x=['Logistic Regression', 'RandomForestClassifier', 'SVM'], y=[lr_acc,rfc_acc,svm_acc], pale
['#BdB76B','#D4AF37','#CE8946'])
plt.xlabel('Classifier Models')
plt.ylabel('Accuracy')
plt.title('Comparison of different models')
```

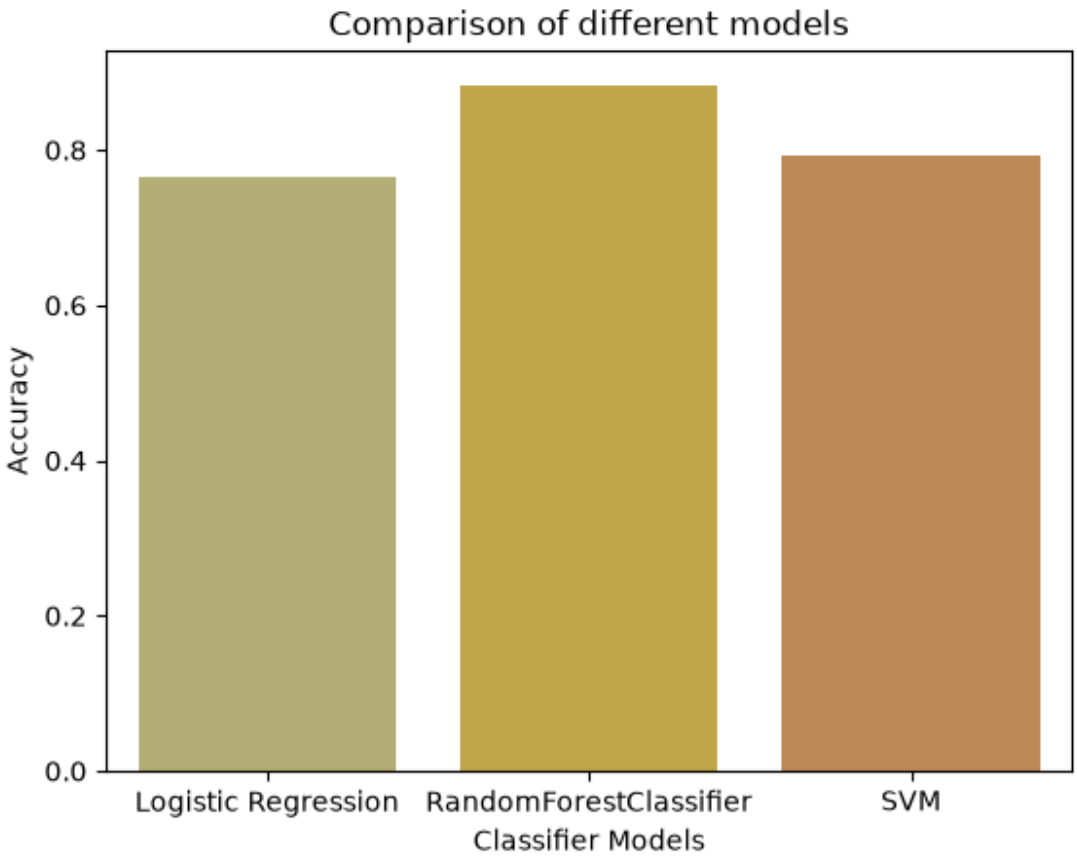


Figure — output of cell [40]

Conclusion

This project successfully completed end-to-end diabetes prediction on the Pima Indian dataset and answered all research questions. The risk of diabetes depends on the following key factors:

1. Glucose, 2. Pregnancies, 3. SkinThickness, 4. Insulin, 5. BMI, 6. DiabetesPedigreeFunction, and 7. Age.

1. Key Findings from EDA (768 records, 65.1% Negative vs 34.9% Positive):

- **Glucose (strongest predictor):** Non-diabetic median < 120, diabetic median > 140. High glucose is a clear indicator of diabetes.
- **BMI:** Non-diabetic patients cluster at 25-35, diabetic patients > 35. Obesity is strongly linked to higher diabetes risk.
- **Insulin:** Non-diabetic centered near 100, diabetic near 200 with wider spread at higher levels. Insulin resistance is a good indicator.
- **SkinThickness:** Median ~20 for non-diabetic vs ~30 for diabetic. Increased skin thickness correlates with diabetes.
- **DiabetesPedigreeFunction:** Non-diabetic mostly 0.25-0.35, diabetic shows increased spread 0.5-1.5. Family history matters.
- **Age:** Majority of patients are 20-30 years, but patients aged 40-55 are more prone to diabetes.
- **Pregnancies:** Higher number of pregnancies shows increased risk of diabetes.
- **BloodPressure:** Median slightly higher for diabetic patients, but distribution overlaps heavily. Not a strong standalone predictor.

2. Model Performance (80/20 stratified split, random_state=42, 154 test samples):

| Model | Accuracy | Correct / 154 | |---|---|---| | Logistic Regression | 76.62% | 118 / 154 | | Random Forest Classifier (n_estimators=100) | **88.31%** | **136 / 154** | | Support Vector Machine (linear kernel) | 79.22% | 122 / 154 |

Best Model: Random Forest Classifier (88.31%) - It outperforms Logistic Regression by ~11.7% and SVM by ~9.1% because it captures non-linear relationships and feature interactions (e.g., Glucose + BMI + Age + DPF) and is more robust to outliers and unscaled features, whereas Logistic Regression and linear SVM assume largely linear boundaries.

3. Limitations:

- Dataset is restricted to Pima Indian females aged 21+, so results may not generalize to other populations.
- Biologically invalid zeros, outliers, feature scaling, class imbalance (SMOTE/class_weight), cross-validation and hyperparameter tuning, and metrics beyond accuracy (Precision, Recall, F1, ROC-AUC, Confusion Matrix) were not fully addressed.

Final Takeaway: Glucose, BMI, Insulin, DPF, SkinThickness, Age and Pregnancies together enable reliable early diabetes prediction, and the Random Forest model at 88.31% accuracy provides a strong baseline for an automated screening aid - not a replacement for medical diagnosis, but a valuable tool for early risk detection.